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Saveetha Institute of Medical And Technical Sciences

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Divide and Conquer Algorithm Comparison

Course Code: CSA0614

Course Name: Design and Analysis of Algorithms

Presented By:

Athavan k(192524036)

Under the supervision:

Dr. A.Sairam

Problem Statement

Algorithm A :

$$T(n) = 2T(n/2) + n$$

Algorithm B :

$$T(n) = 2T(n/2) + n^2$$

Objective:

- Understand Divide and Conquer strategy.
- Apply Master Theorem.
- Determine time complexity.
- Compare recursive structures.
- Identify the more efficient algorithm.

The Divide-and-Conquer

Divide:

Split the problem of size n into 2 subproblems, each of size n/2.

Conquer:

Recursively solve each subproblem using the same algorithm.

Combine:

Merge the subproblem results — this step's cost is where the two algorithms differ.

Two Algorithms Under Comparison

ALGORITHM A

Linear-Cost Combine (e.g., Merge Sort style)

$T(n) = 2T(n/2) + n$

Splits input into two halves, recurses, then combines results in linear time — e.g., merging two sorted sub-arrays element by element.

ALGORITHM B

Quadratic-Cost Combine (all-pairs style)

$T(n) = 2T(n/2) + n^2$

Splits input into two halves, recurses, then combines results by comparing every element pair across the two halves — an $O(n^2)$ combine step.

Pseudocode for Both Algorithms

ALGORITHM A — $T(n) = 2T(n/2) + n$

```
ALGO_A(arr, low, high):
  if low >= high: return
  mid = (low + high) / 2
  ALGO_A(arr, low, mid)      // T(n/2)
  ALGO_A(arr, mid+1, high)   // T(n/2)
  LINEAR_COMBINE(arr, low,
                 mid, high)  // O(n)
```

ALGORITHM B — $T(n) = 2T(n/2) + n^2$

```
ALGO_B(arr, low, high):
  if low >= high: return
  mid = (low + high) / 2
  ALGO_B(arr, low, mid)        // T(n/2)
  ALGO_B(arr, mid+1, high)     // T(n/2)
  for i in left_half:
    for j in right_half:
      COMPARE(i, j)            // O(n²)
```

Dry Run Recursion Tree for n = 8

Algorithm A: $T(n) = 2T(n/2) + n$

Level	Subproblems	Size	Cost/Level
0	1	8	8
1	2	4	8
2	4	2	8
3	8	1	8

Total ≈ $8 \times (\log_2 8 + 1) = 8 \times 4 \rightarrow \Theta(n \log n)$

Cost is evenly spread across all levels — no single level dominates.

Algorithm B: $T(n) = 2T(n/2) + n^2$

Level	Subproblems	Size	Cost/Level
0	1	8	64
1	2	4	32
2	4	2	16
3	8	1	8

Total = 64 + 32 + 16 + 8 = 120 → root level alone ≈ 53% of total cost

Cost shrinks geometrically going down the tree — the root (top) level dominates → $\Theta(n^2)$.

Applying the Master Theorem

Master Theorem: $T(n) = aT(n/b) + f(n)$ — compare $f(n)$ with $n^{(\log_m b\ a)}$

Algorithm A — $T(n) = 2T(n/2) + n$

$a = 2,\ b = 2,\ f(n) = n$

$n^{(\log_2 2)} = n^1 = n$

Compare: $f(n) = \Theta(n^{(\log_2 2)})$

→ $f(n)$ and $n^{(\log_2 2)}$ grow at the same rate

→ CASE 2 of Master Theorem applies

$T(n) = \Theta(n \log n)$

Algorithm B — $T(n) = 2T(n/2) + n^2$

$a = 2,\ b = 2,\ f(n) = n^2$

$n^{(\log_2 2)} = n^1 = n$

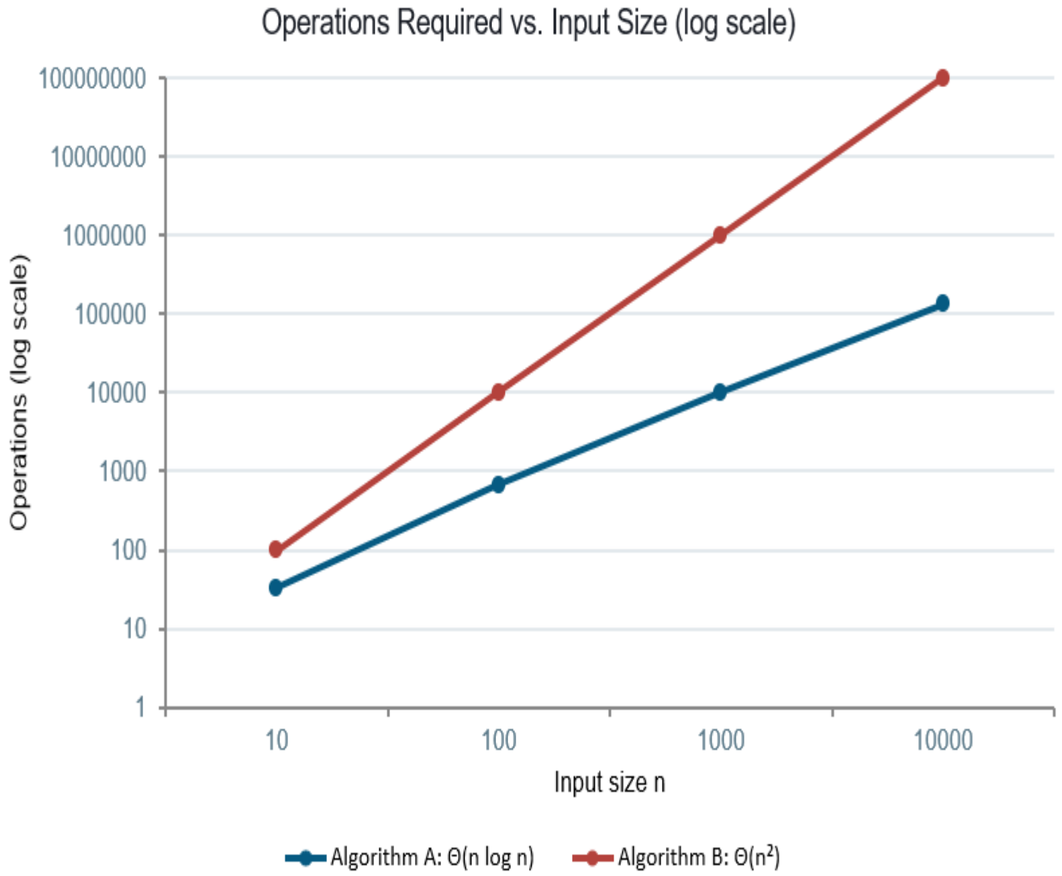
Compare: $f(n) = n^2 = \Omega(n^{(1+\varepsilon)})$ for $\varepsilon = 1$

Regularity check: $2\cdot(n/2)^2 = n^2/2 \le c\cdot n^2$ for $c = 1/2 < 1$ ✓

→ CASE 3 of Master Theorem applies

$T(n) = \Theta(n^2)$

Growth Rate: Θ(n log n) vs. Θ(n²)



Operation Count by Input Size

n	n log n	n²
10	33	100
100	664	10,000
1000	9,966	1,000,000
10000	132,877	100,000,000

At n = 10,000:

Algorithm A needs ≈ 132,877 ops

Algorithm B needs ≈ 100,000,000 ops

Algorithm B requires roughly 750× more operations than Algorithm A at this scale.

Where Each Combine Strategy Is Used

Algorithm A — $\Theta(n \log n)$

- Merge Sort — linear merge of two sorted halves
- External sorting for large disk-resident datasets
- Parallel / distributed sort-and-merge pipelines
- Inversion counting during a linear merge pass

Algorithm B — $\Theta(n^2)$

- Brute-force closest-pair combine across a dividing line
- Naive cross-comparison of two point sets
- Educational baseline before optimizing to O(n log n)
- Small-n subroutines where n^2 combine cost is negligible

Findings and Interpretation

1

The recurring subproblem term is identical

Both algorithms split into a = 2 subproblems of size n/2 — the difference lies entirely in the combine cost f(n).

2

Combine cost determines the Master Theorem case

A linear $f(n) = n$ keeps pace with the recursion (Case 2); a quadratic $f(n) = n^2$ overwhelms it (Case 3).

3

Root-heavy vs. balanced work distribution

Algorithm B's cost concentrates at the top level of the tree; Algorithm A spreads cost evenly across log n levels.

4

Algorithm A scales far better for large n

The gap between $\Theta(n \log n)$ and $\Theta(n^2)$ widens rapidly — at n = 10,000 the gap is already ≈750×.

Conclusion

- Although both algorithms divide the problem into the same number of subproblems, the combine cost significantly affects their overall performance.
- Algorithm A is more efficient and scalable because its linear combine cost results in a lower time complexity.
- Algorithm B is less efficient for large input sizes due to its quadratic combine cost.
- Overall, this comparison demonstrates that minimizing the combine cost is essential for designing efficient divide-and-conquer algorithms, making Algorithm A the preferred choice for large-scale applications.

Reference

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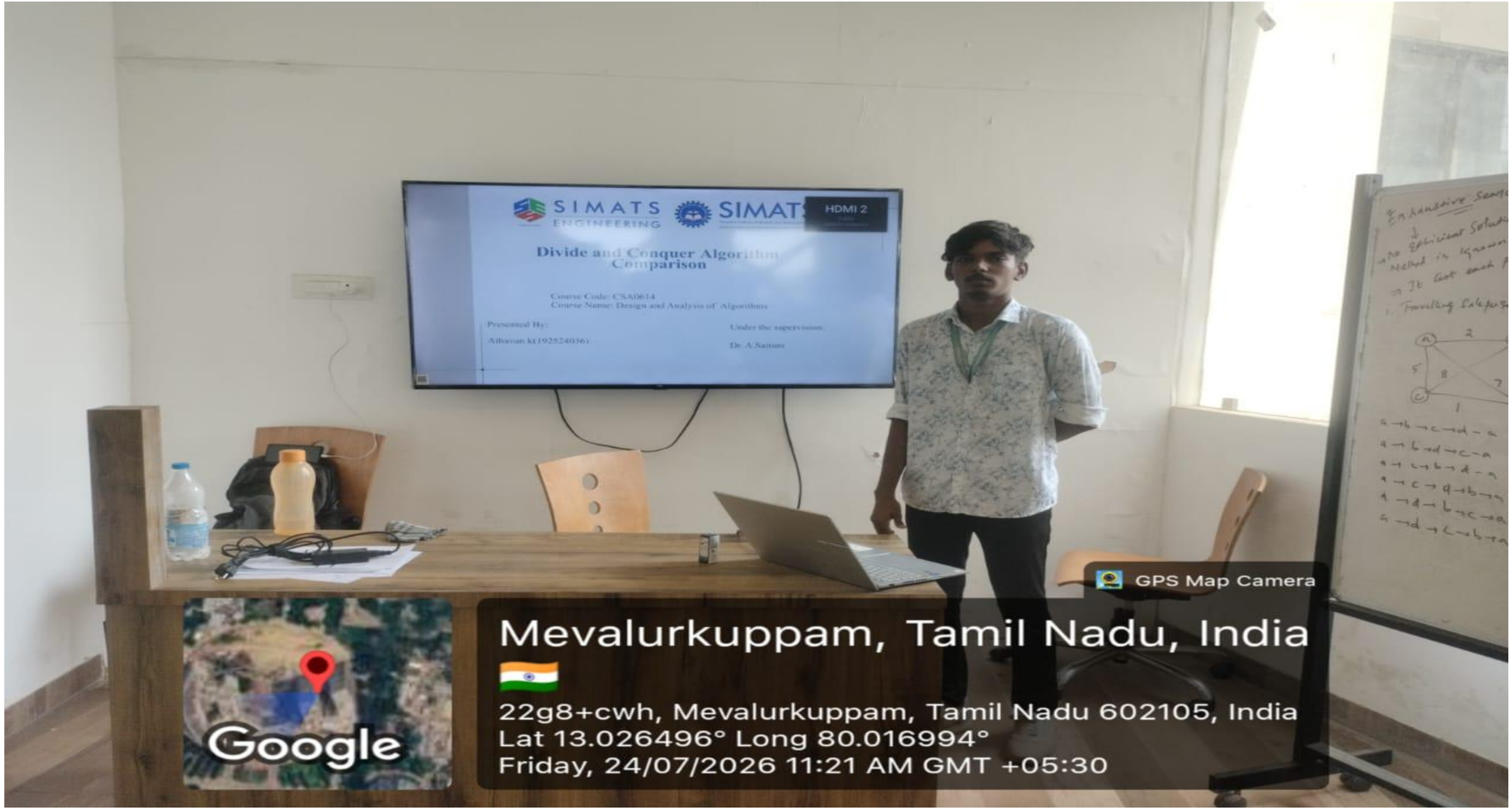
THANK YOU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Course Name: Design and Analysis of Algorithms
Presented By:
Athavan k(192524036)
Under the supervision:
Dr. A.Saitsun
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
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